Natanael S. Alpay

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SUMMARY

Mathematics Ph.D. candidate with 10+ publications and preprints in analysis, probability, and machine learning. Uses LLMs as conjecture engines, then verifies candidate results through mathematical proof and reproducible computation. Applied work spans healthcare signal prioritization and ongoing research focused on developing and evaluating neural-network based hedging strategies. Has presented over 20 research talks. Seek quantitative research and machine-learning role

EDUCATION

Ph.D. Mathematics	University of California, Irvine	**Expected 2028**
M.S. Applied Mathematics	University of California, Irvine	**2025**
B.S. Mathematics	Chapman University	**2022**
B.S. Physics	Chapman University	**2022**
B.S. Computer Science	Chapman University	**2022**
B.A. French	Chapman University	**2022**

- Earned the Ronald M. Huntington Outstanding Scholarship Award and DataFest First Place/Best Insight.

SKILLS

- Programming: Python, SQL, R, C/C++, JavaScript, MATLAB, Mathematica.
- Python Libraries: PyTorch, NumPy, pandas, Matplotlib.
- Machine Learning: Neural networks, SGD, logistic regression, gradient boosting, temporal validation, calibration.
- Quantitative Methods: Stochastic processes, optimization, Monte Carlo simulation, numerical analysis, kernel methods.
- Languages: English, French, Russian, Hebrew.

SELECTED PROJECTS

Deep Bellman Hedging and Portfolio Compression **2026 - Present**

- Implemented Python/PyTorch neural networks that choose stock holdings to offset option losses after trading costs; compared two ways of penalizing large losses (CVaR and entropic risk).
- Evaluated strategies on historical SPY data and Monte Carlo stock-price simulations (Black-Scholes), using separate training, validation, and test sets; compared with a standard hedge (delta hedging) and no hedge.

Complex Optimization and Kernel Regression **2026**

- Developed Python implementations of complex-valued gradient methods to reconstruct special solutions and study which solutions the algorithms learn. Examples of inverse problems with initial conditions arising from quantum mechanics.
- Established convergence guarantees under stated assumptions and checked them through numerical experiments.

Machine Learning for Drug and Vaccine Safety **2025 - 2026**

- Prioritized VAERS vaccine-symptom pairs for review using Python implementations of logistic regression and gradient boosting; evaluated next-quarter predictions against statistical signal scores. Presented at CHOC's Pediatric and Lifespan Data Science Conference (April 29, 2026).
- Refined Python gradient-boosting models to prioritize FAERS drug-safety signals, improving ranking performance (AUPRC) from approximately 0.68 to 0.85 on a pilot holdout of future quarters.

RESEARCH AND TEACHING EXPERIENCE

University of California, Irvine **Sep 2022 - Present**

Graduate Research and Teaching Assistant

- Investigate simpler portfolio summaries for managing risk through Deep Bellman Hedging, a current research direction.
- Proved tight bounds on how functions fluctuate across scales, with calculations implemented in Python; tested ideas suggested by large language models through proofs and computational checks.
- Taught Numerical Analysis I and II as instructor of record (Summer 2026), supported mathematics courses as a teaching assistant, and mentored DATA@UCI students in Python analysis and presentations (Jan-Jun 2025).
- Lead the Deep Hedging Directed Reading Program, guiding undergraduates through research papers, Python experiments, and critical evaluation.
- Mentored eight undergraduates on two random-matrix projects, guiding Python experiments and two research posters.

Chapman University **Jun 2020 - Sep 2022**

Machine Learning Research and Data Analysis

- Analyzed Brain Institute data using machine-learning models in Python and analysis tools in SQL and JavaScript.

SELECTED PUBLICATIONS AND PREPRINTS

Alpay & Battaglia. Complex Stochastic Gradient Descent and Directional Bias in Reproducing Kernel Hilbert Spaces. arXiv, 2026.
Alpay & Ivanisvili. Sharp Lower Bounds for Dyadic Square Functions of Indicator Functions of Sets. arXiv, 2025; revised 2026.
Alpay, Cerejeiras & Kähler. Generalized Fock-space and fractional derivatives: Uniqueness of sampling and interpolation sets. Results in Applied Mathematics, 2026.

Interests: Weightlifting, LEGO, karaoke.